

# FDDT Case Report

Fraud Document Detection Tool — system-assisted analysis for human review. See Section 6 for what this report is and is not.

## 1. Case Details

|                   |                                                              |
|-------------------|--------------------------------------------------------------|
| Case ID           | CASE-E29DD0A6    e29dd0a6-1658-4961-93dd-cafa96440333        |
| Submitted by      | testreviewer L!                                              |
| Submission date   | 2026-10-07 06:44 UTC                                         |
| Documents         | 1    Types: School document                                  |
| Case type         | School document                                              |
| Status / decision | Submitted / <b>Pending review</b>                            |
| Report generated  | 2026-10-07 06:45 UTC by <b>testreviewer L!</b> (Reviewer L1) |
| Report ID         | f4713e96-e946-498d-8662-1e484b3a272b                         |

**Preliminary report.** Automated analysis had not finished when this report was generated. Still outstanding: Field validation of 'case2.pdf'; Issuer verification of 'case2.pdf'; Error level analysis of 'case2.pdf'; Copy move detection of 'case2.pdf'; Duplicate detection of 'case2.pdf'.

## 2. Executive Summary

**Not yet scored.** The risk engine has not produced an assessment for this case, so there is no overall tier.

**Why:** No risk tier yet — automated analysis is still in progress.

**Decision status:** Pending review.

**Exceptions:** 2 item(s) did not pass, 2 of them shown as highlighted regions on the source pages. Section 5 lists each one; Section 9 shows them on the documents.

This summary is deliberately short. The reasons behind the tier are in Section 3, the complete checklist is in Section 4.

### 3. Explainable Findings

The case has not been scored, so there are no triggered risk reasons yet.

## 4. All Checks

The completeness record: every check that ran on every document, including those that passed. Section 3 is the narrative; this is the checklist.

Document 1 — case2.pdf (School document · 3 passed, 1 flagged)

| Check                                                                                                                                             | Result | Details                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Metadata forensics</b><br>PDF metadata, edit history and incremental-save markers.                                                             | Pass   | No anomaly detected by this check.                                                                                                                                                                                                                                    |
| <b>Font consistency</b><br>Text set in a different font family from the text around it (text layer, or OCR font recognition on scans).            | Pass   | No anomaly detected by this check.                                                                                                                                                                                                                                    |
| <b>Visual inconsistency / AI-generation review</b><br>Model review of fonts, alignment, contrast, sharpness, lighting and signs of AI generation. | Flag   | 2 finding(s). Page 1 — Text alignment: The text in the bank details table is misaligned; some lines are slightly shifted vertically relative to others, especially in the middle rows (around 0.3 to 0.45 vertical fraction). This is a vision-model judgment, not... |
| <b>Signature / stamp detection</b><br>Whether a signature or stamp mark is present where this kind of document normally carries one.              | Pass   | 3 signature/stamp region(s) located (presence only).                                                                                                                                                                                                                  |

Linked cases (reviewer note, not scored)

Other cases of this company that share a scanning device, a parent/guardian or a family/student ID with this one. Shared office scanners are common, so this is context for the reviewer only; it does not affect the risk score.

| Case          | Shared                                                                                                                                                                                        |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CASE-11953D10 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Ahmed Salem Alkindi (father's name in HAMDA AHMED SALEM ALKINDI)') |
| CASE-11DD9B6A | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Mr. Ahmed Salem Abdulla AlKindi'); same family number (1245)       |
| CASE-37B33CBB | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Ahmed Salem Alkindi (father's name in HAMDA AHMED SALEM ALKINDI)') |
| CASE-3B02E9B2 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Ahmed Salem Alkindi (father's name in HAMDA AHMED SALEM ALKINDI)') |
| CASE-6CB8F7E1 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Ahmed Salem Alkindi (father's name in HAMDA AHMED SALEM ALKINDI)') |
| CASE-6CE43E0C | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Ahmed Salem Alkindi (father's name in HAMDA AHMED SALEM ALKINDI)') |
| CASE-7978A329 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Mr. Ahmed Salem Abdulla AlKindi'); same family number (1245)       |
| CASE-92AA4164 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Ahmed Salem Alkindi (father's name in HAMDA AHMED SALEM ALKINDI)') |
| CASE-942C8493 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Ahmed Salem Alkindi (father's name in HAMDA AHMED SALEM ALKINDI)') |
| CASE-A7705426 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Mr. Ahmed Salem Abdulla AlKindi'); same family number (1245)       |
| CASE-BDFC21C5 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Mr. Ahmed Salem Abdulla AlKindi'); same family number (1245)       |

|               |                                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CASE-CF609781 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Mr. Ahmed Salem Abdulla AlKindi'); same family number (1245)       |
| CASE-DB43653A | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Ahmed Salem Alkindi (father's name in HAMDA AHMED SALEM ALKINDI)') |
| CASE-DE627FA2 | same scanning device (PPS - SFO2-A-MFP01, serial 00206BE9FB63); same parent/guardian ('Mr. Ahmed Salem Abdulla AlKindi' / 'Mr. Ahmed Salem Abdulla AlKindi'); same family number (1245)       |

## 5. Exceptions

Everything that did **not** pass, drawn from Section 4 and written as concrete items (2 in total). A region ID (R1, R2, ...) points to the highlight that shows the exception on the source page in Section 9 — click it to jump there.

| No. | Exception                                                                                                                                                                                                                                                                                                                                                                                                                     | Where                           | Highlight |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------|
| 1   | <b>Visual review: text alignment inconsistency (model judgment)</b><br>Page 1 — Text alignment: The text in the bank details table is misaligned; some lines are slightly shifted vertically relative to others, especially in the middle rows (around 0.3 to 0.45 vertical fraction). This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding.                          | Document 1 (case2.pdf) · page 1 | R1        |
| 2   | <b>Visual review: resolution / sharpness consistency inconsistency (model judgment)</b><br>Page 1 — Resolution / sharpness consistency: The handwritten signatures at the bottom are noticeably softer and blurrier compared to the sharp printed text. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding. Not counted: it is about the detected signature (cov... | Document 1 (case2.pdf) · page 1 | R2        |

## 6. Limitations and Assumptions

Please read this section before relying on anything in this report.

### Limitations of This Analysis

#### What this report is

This is **system-assisted analysis to support human review**. It is not a certified forensic or legal determination that any document is genuine, forged or fraudulent, and it must not be presented as one.

#### Three kinds of finding

**Deterministic pixel and structure checks** — error level analysis, copy-move detection and PDF metadata forensics — are computed by fixed algorithms over the file’s pixels and structure. Localized findings are drawn with **solid** boxes (ELA in red, copy-move in orange). They are signals, not proof: ELA in particular can false-positive on legitimately recompressed scans and can miss print-and-rescan tampering.

**Deterministic rule-based field checks** — field validation and cross-document consistency — compare extracted values by fixed rules and are drawn with **solid purple** boxes. They are exact about the values they compare, but only as reliable as the OCR and extraction that produced those values.

**Model-judgment checks** — the visual inconsistency review, the AI-generation assessment and the signature/stamp comparison — are **probabilistic assessments** made by a vision-language model, not deterministic proof. They can vary from run to run, and their boxes are only approximate; they are the **dashed** boxes captioned “approximate” throughout this report. Signature and stamp detection shows presence and placement only and is not an identity match; the AI-generation assessment is experimental, and its accuracy on scanned business documents has not been validated.

#### What the system can and cannot see

The system evaluates the **submitted document file itself**. It cannot verify facts external to the document — for example whether a real-world transaction actually occurred, whether the named issuer really produced the document, or (without a pre-existing reference signature) who a signatory is. Printed Arabic and English text are read; handwritten Arabic is not reliably read.

#### Absence of a finding

**The absence of a flagged finding does not guarantee that a document is genuine.** It means only that no supported check detected an anomaly.

#### Check coverage on this case

Not every check applies to every document or case. This is what actually ran here:

| Check                                       | Status         | Coverage on this case                                       |
|---------------------------------------------|----------------|-------------------------------------------------------------|
| Metadata forensics                          | Ran            | Ran on 1 of 1 document(s); 0 flagged.                       |
| Font consistency                            | Ran            | Ran on 1 of 1 document(s); 0 flagged.                       |
| Error level analysis (ELA)                  | Did not run    | Did not run on any of the 1 document(s).                    |
| Copy-move detection                         | Did not run    | Did not run on any of the 1 document(s).                    |
| Visual inconsistency / AI-generation review | Ran            | Ran on 1 of 1 document(s); 1 flagged.                       |
| Signature / stamp detection                 | Ran            | Ran on 1 of 1 document(s); 0 flagged.                       |
| Duplicate detection                         | Did not run    | Did not run on any of the 1 document(s).                    |
| Field validation                            | Did not run    | Did not run on any of the 1 document(s).                    |
| Issuer verification                         | Did not run    | Did not run on any of the 1 document(s).                    |
| Signature comparison (advisory)             | Did not run    | Did not run — no reference signature was set for this case. |
| Cross-document consistency                  | Not applicable | Not applicable — this case has a single document.           |

#### Risk score and rule versions

This case had not been scored when the report was generated, so no risk-rule versions apply.

### Assumptions

What the analysis took to be true without independently verifying it. These are specific to what ran on this case.

- Field values were read by OCR followed by language-model extraction and are assumed to have been read correctly. Each field carries a confidence and an “uncertain” flag. This build records that flag but does not automatically route low-confidence documents to manual review, so it is listed here for the reviewer instead: no field was marked uncertain on this case.
- The SHA-256 chain of custody shows the stored file is unchanged since upload. It does not show that the uploaded file is the issuer’s original — a document altered before upload is judged on its content, like any other.
- Risk-rule weights, severities and tier thresholds are administrator-configured and assumed to be appropriately calibrated; the score is advisory and is not a probability of fraud.



## 7. Audit Trail

Chronological case-level events, read from the append-only audit log (times in UTC). This report does not write to or duplicate the log.

| Time                 | Event                                                             | Actor                            | Detail                                    |
|----------------------|-------------------------------------------------------------------|----------------------------------|-------------------------------------------|
| 2026-10-07 06:44 UTC | Case created                                                      | testreviewer L!<br>(Reviewer L1) |                                           |
| 2026-10-07 06:44 UTC | Document uploaded                                                 | testreviewer L!<br>(Reviewer L1) | case2.pdf; SHA-256<br>e9c3ea682c7407ba... |
| 2026-10-07 06:44 UTC | Metadata forensics completed                                      | System                           | case2.pdf                                 |
| 2026-10-07 06:44 UTC | Signature / stamp detection completed                             | System                           | case2.pdf                                 |
| 2026-10-07 06:45 UTC | Visual inconsistency review completed                             | System                           | case2.pdf                                 |
| 2026-10-07 06:45 UTC | Font consistency check completed                                  | System                           | case2.pdf                                 |
| 2026-10-07 06:45 UTC | Text extraction, classification and field<br>extraction completed | System                           | case2.pdf                                 |

## 8. Appendix

The detailed technical record, for someone auditing the analysis itself rather than a first-pass reviewer.

### 8.1 Extracted fields

Normalized values as stored at extraction time. “Located” is the page the value was matched to (used for the purple highlights); a blank means no position was recorded. Arabic values are shown right-to-left, numbers left-to-right.

#### Document 1 — case2.pdf

| Field                      | Value                                                                     | Confidence | Located |
|----------------------------|---------------------------------------------------------------------------|------------|---------|
| Issuer                     | THE GULF INTERNATIONAL PRIVATE أكاديمية الخليج الدولية الخاصة<br>ACADEMYO | 99%        | p.1     |
| Reference number           | 52086                                                                     | 98%        | p.1     |
| Date                       | 2023-07-05                                                                | 99%        | p.1     |
| Amount                     | 57,182.50                                                                 | 98%        | p.1     |
| Subtotal                   | Not found                                                                 |            |         |
| Tax rate                   | 5.00                                                                      | 98%        | p.1     |
| Tax amount                 | 42.50                                                                     | 98%        | p.1     |
| Student name               | GHAYA AHMED SALEM ABDULLA ALKINDI                                         | 99%        | p.1     |
| Grade                      | KG2                                                                       | 99%        | p.1     |
| Parent name                | Mr. Ahmed Salem Abdulla AlKindi                                           | 95%        | p.1     |
| Family number              | 1245                                                                      | 95%        | p.1     |
| Beneficiary name           | The Gulf International Private Academy                                    | 98%        | p.1     |
| Beneficiary bank name      | Abu Dhabi Commercial Bank (ADCB)                                          | 98%        | p.1     |
| Beneficiary account number | 260337020001                                                              | 98%        | p.1     |
| Beneficiary iban number    | AE250030000260337020001                                                   | 98%        | p.1     |
| Beneficiary branch name    | Khaleefa Street, Al Ain, Abu Dhabi                                        | 98%        | p.1     |
| Email                      | Accountant@gipa.ae                                                        | 98%        | p.1     |

### 8.2 Document hashes (chain of custody)

SHA-256 of each original as recorded when it was uploaded to immutable storage, re-checked against the stored file when this report was generated.

| Document                  | SHA-256                                                          | Verification |
|---------------------------|------------------------------------------------------------------|--------------|
| case2.pdf<br>59,415 bytes | e9c3ea682c7407ba50d8564f90fb668ee106280583b785e4f80ea151456a70b6 | VERIFIED     |

### 8.3 Technical parameters used by each check

#### Page rendering (input to ELA, copy-move, duplicate, visual review)

|                   |                                  |
|-------------------|----------------------------------|
| Render resolution | 200 DPI                          |
| Colour model      | RGB, converted to BGR for OpenCV |

#### Metadata forensics

|                                |                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------|
| Libraries                      | pikepdf / PyMuPDF (PDF structure, XMP, incremental updates)                          |
| Modified-after-created rule    | flagged if ModDate is more than 1 second(s) after CreationDate                       |
| Editing-software names matched | photoshop, gimp, illustrator, affinity, coreldraw, paint.net, ptxlr, canva, inkscape |

#### Visual inconsistency / AI-generation review

|                           |                                                             |
|---------------------------|-------------------------------------------------------------|
| Model                     | gpt-4.1-mini                                                |
| Independent runs per page | 2                                                           |
| A finding is kept if      | both runs report it, or one reports it with high confidence |
| Image sent to model       | page render, long edge ≤ 1536px                             |

Signature / stamp detection and comparison

|                             |                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------|
| Model                       | gpt-4.1-mini                                                                            |
| Pages analysed per document | first 10                                                                                |
| Output                      | presence and placement only; comparison is a qualitative verdict, never a numeric score |

Field locations (purple highlights)

|                 |                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------|
| Method          | extracted values matched back onto OCR word positions (Azure Document Intelligence Layout) |
| Repeated values | resolved by nearby keywords (“total”, “subtotal”, “VAT”, “date”...)                        |
| Accuracy        | approximate; a field that cannot be matched is reported without a highlight                |

## 9. PDF Highlighted Regions

Every flagged page, rendered from the original with its findings drawn in. These are copies made for this report; the original files are untouched. Only pages with at least one highlight appear. Each highlight has a region ID that Section 5 refers back to.

### Colour key

|                                                                                   |                                                 |
|-----------------------------------------------------------------------------------|-------------------------------------------------|
|  | ELA (solid red)                                 |
|  | Copy-move (solid orange)                        |
|  | Font mismatch in the text layer (solid fuchsia) |
|  | Field exception (solid purple)                  |
|  | Model judgment (dashed blue, approximate)       |
|  | Page-level finding (not localized)              |

### Region index

| Region | Document               | Page | What it shows                                                    |
|--------|------------------------|------|------------------------------------------------------------------|
| R1     | Document 1 (case2.pdf) | 1    | Visual review — text alignment (approximate)                     |
| R2     | Document 1 (case2.pdf) | 1    | Visual review — resolution / sharpness consistency (approximate) |

## case2.pdf — page 1

Section 9 · Document 1 of 1 · regions R1, R2

 Model judgment (dashed blue, approximate)

**TAX INVOICE**

Inv. : 52086  
TRN : 100216382000003  
Date : 05/07/2023

The Gulf International Private Academy in Al Ain certifies that the below student(s) is/are registered in our school for the academic year 2023-2024. Fees for registered students(s) are as follows:

R1 · Visual review — text alignment (approximate)

| Name                              | Grade | Tuition Fees | Books | Uniform | Technology Fee | Transport | 5% Vat | Total Fees |
|-----------------------------------|-------|--------------|-------|---------|----------------|-----------|--------|------------|
| GHAYA AHMED SALEM ABDULLA ALKINDI | KG2   | 52,130       | 650   | 300     | 550            | 3,510     | 42.50  | 57,182.50  |

Bank Detail:

|                             |                                        |
|-----------------------------|----------------------------------------|
| Beneficiary Name:           | The Gulf International Private Academy |
| Beneficiary Bank Name:      | Abu Dhabi Commercial Bank (ADCB)       |
| Beneficiary Account Number: | 260337020001                           |
| Beneficiary IBAN Number:    | AE250030000260337020001                |
| Beneficiary Branch Name:    | Khalifa Street, Al Ain, Abu Dhabi      |
| New Email ID:               | Accountant@giipa.ae                    |

This letter has been issued upon the guardian's request without any liabilities for our school. The fees stated in this document are in accordance to the Abu Dhabi Department of Education and Knowledge (ADEK) approved school fees.

Parent Name : Mr. Ahmed Salem Abdulla AlKindi  
Family Number : 1245

Registration Department      Accounts Department

R2 · Visual review — resolution / sharpness consistency (approximate)

Potential Crows Here

مدفوق برود: 52086، الفاتحة: 100216382000003، التاريخ: 05/07/2023، هاتف: +971 3 767 6905، فاكس: +971 3 767 6908، بريد: info@giipa.ae  
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**R1. Visual inconsistency / AI-generation review**

Page 1 — Text alignment: The text in the bank details table is misaligned; some lines are slightly shifted vertically relative to others, especially in the middle rows (around 0.3 to 0.45 vertical fraction). This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding.

**R2. Visual inconsistency / AI-generation review**

Page 1 — Resolution / sharpness consistency: The handwritten signatures at the bottom are noticeably softer and blurrier compared to the sharp printed text. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding. Not counted: it is about the detected signature (covers 100% of it) — pen strokes and rubber-stamp ink are naturally softer than printed text.